

PANEER SELVAM V

[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

☎ +91 7094395837 ✉ paneerselvamv4@gmail.com

Skills

- **Languages** - JavaScript (ES6+), TypeScript
- **Frontend** - React.js, Next.js, React Components, Props, State, Hooks
- **Backend** - Node.js, Express.js, RESTful APIs, JSON, Asynchronous Programming
- **Databases** - MongoDB, PostgreSQL, Astra DB, Mongoose, Prisma ORM
- **Devops** - AWS, Docker, CI/CD
- **Tools & Deployment** - Git, Github, npm, Postman, Swagger/OpenAPI, Browser DevTools, Vercel, Netlify, Render
- **Security & API Development** - API Validation, Middleware, Authentication, Error Handling, Helmet, Rate Limiting
- **AI Engineering** - LLM integration, Vector Databases, Embeddings, RAG Architectures, LangChain, Prompt Engineering
- **AI Tools & Technologies** - Ollama, Cursor, Claude, OpenAI, DeepSeek, MCP Fundamentals
- **Core CS** - DSA, OOP, Problem Solving, Design Patterns

Projects

1. *Ben10 Alien Explorer*

- Developed a full-stack AI-powered application using Next.js and TypeScript, featuring searchable alien directories, series filtering, detailed pages, and authenticated user workflows.
- Built and **tuned a domain-specific RAG assistant** using Gemini embeddings, **Astra DB vector search**, and streamed Gemini responses with retrieved context.
- Optimized the assistant's **retrieval context and prompting** to improve the relevance and quality of generated responses.
- Implemented PostgreSQL persistence with Prisma for alien records, favorites, user accounts, and chat history.
- Built authenticated API routes with Zod request validation, relational constraints, and per-user rate limiting.
- **Tech - Next.js | TypeScript | PostgreSQL | Prisma | Astra DB | LangChain | Framer Motion**

2. *Ben10 Aliens REST API*

- Developed a versioned REST API for retrieving and creating alien data with resource-based endpoints, series routing, and query-parameter filtering.
- Integrated MongoDB with Mongoose for schema-based persistence and asynchronous database operations.
- Implemented request-validation middleware, HTTP status-code handling, controller-level error handling, CORS, Helmet security headers, and request-rate limiting.
- Created and executed Postman collections and tests to validate API endpoints, request/response behavior, and error scenarios; documented the API with Swagger/OpenAPI and deployed it on Render.
- **Tech - Node.js | Express.js | MongoDB | Mongoose | REST APIs | Swagger/OpenAPI | Postman | Render**

3. *Custom Promise Library-(PizzaHub Visualization)*

- Implemented a custom JavaScript Promise library to explore Promise states, chaining, asynchronous execution, and rejection handling.
- Built a PizzaHub ordering visualizer that connects asynchronous order processing with visible lifecycle stages.
- Separated order processing from UI logic and implemented cancellation handling for interrupted workflows.
- **Tech - JavaScript | Promises | Asynchronous Programming**

Education

Hindusthan college of Arts and Science

Bachelor of Science - Computer Science, CGPA – 7.4

2022 – 2025